

Quantum Spatial Mapping and Quantum Teleportation

A plain guide to qubits, entanglement, teleportation, and the idea of mapping space with quantum information

Contents

Preface	1
Chapter 1. Maps and Messages	2
Chapter 2. The Qubit	5
Chapter 3. Entanglement	8
Chapter 4. The Copying Ban	11
Chapter 5. How Teleportation Actually Works	13
Chapter 6. Teleportation in the Laboratory	17
Chapter 7. Limits: Why This Is Not a Transporter	20
Chapter 8. Mapping Space With Quantum Information	23
Chapter 9. Where the Two Ideas Meet	27
Chapter 10. The Quantum Spatial Mapping Project	30
Closing	35
Further Reading	36

Preface

This book is about two ideas that are often mentioned in the same breath but are not the same thing. The first, quantum teleportation, is established physics: a precisely defined protocol, published in 1993 and demonstrated in many laboratories since, that transfers the quantum state of one system onto another using shared entanglement and an ordinary message. Everything said about qubits, superposition, entanglement, the no-cloning theorem, Bell tests and the teleportation protocol in these pages reflects mainstream, textbook physics, and the named experiments are real. The second idea, which I shall call quantum spatial mapping, is not an established field. Parts of it touch real research, such as quantum sensing, quantum reference frames and the study of particles moving in curved space, and I describe those as they stand. Other parts are interpretation or speculation, and wherever the discussion moves beyond what has been measured or proved I say so plainly, usually under a heading that marks it as speculative. The final chapter discusses the Quantum Spatial Mapping website, which describes itself as a playful, evolving proposal; I quote only what that page and its accompanying formula sheet actually say, and I treat the project as the open invitation to discussion it presents itself as, not as a finished theory or a laboratory result.

Chapter 1. Maps and Messages

Imagine two friends, one in Halifax and one in Edinburgh, who want to share something. If it is a parcel, it goes in a plane. If it is a message, it can go by telephone, by email, or written on a postcard. If it is a description of something, say the precise colour of a sunset, the friend in Halifax has to decide how to describe it, and the friend in Edinburgh has to rebuild an impression from that description. In everyday life we hardly notice the difference between sending a thing, sending a message, and sending a description that lets someone rebuild a thing. Quantum physics forces us to notice, because at the quantum scale the three behave very differently.

Quantum teleportation lives in the space between those three. It does not send a thing, in the sense of an object made of matter. It does not send a complete description either, because, as we shall see, a complete description of a single quantum state cannot be read off it. What it does is stranger and more modest: it uses a pre-arranged pair of linked particles, together with a short ordinary message, to make a particle at the far end take on the exact quantum state that a particle at the near end used to have. The original state is destroyed in the process. Nothing travels faster than light. Nobody beams up.

A map is a different sort of object altogether. A map does not move anything; it records relationships. It tells you that the harbour is to the east of the church and that the hill is a hundred metres high. Good maps depend on good measurements, a shared idea of which way is north, a common unit of length, and a way of comparing readings taken at different places and times. When people talk about mapping space with quantum information, they might mean several different things. They might mean using quantum systems as extraordinarily sensitive measuring instruments, which is real and active technology. They might mean the question of how two distant parties agree on directions and clocks when their instruments are themselves quantum objects, which is a real research topic with a respectable literature. Or they might mean the much more ambitious and

speculative idea that space itself is built out of, or mapped by, patterns of quantum entanglement. That last idea is taken seriously by some theoretical physicists working on quantum gravity, but it is unproven and remains a matter of active research.

The reason teleportation and mapping are worth discussing together is that both rest on the same peculiar resource: entanglement, a kind of correlation between quantum systems that persists however far apart they are. Teleportation spends entanglement to move a state. Mapping, in the various senses above, uses correlations to compare, to measure, or to describe relationships across distance. Both run into the same limits. Neither lets you send a message faster than light. Both need an ordinary classical channel at some point. Both are fragile, because quantum correlations leak away into the environment with alarming speed. And both depend, in a quiet way that is easy to overlook, on the two ends agreeing about frames of reference: which direction is up, which way a measuring device is turned, when a clock started.

This book sets out to explain these ideas without equations doing the heavy lifting, though a few appear where they genuinely help. We begin with the qubit, the basic unit of quantum information, and with superposition, the feature of quantum systems that makes them different from switches and coins. We then meet entanglement and the experiments, beginning with those of Alain Aspect and his colleagues in the early 1980s, that showed it to be real. We look at the no-cloning theorem, which explains why quantum information cannot simply be photocopied, and which is the reason teleportation has to work the way it does. Then we go through the teleportation protocol step by step, look at how it has been done in practice, and take an honest account of its limits.

The second half of the book turns to space. We look at what quantum systems can and cannot tell us about where things are and how they are arranged, at the role of reference frames, and at the places where the language of teleportation and the language of mapping genuinely meet. Finally, we turn to the Quantum Spatial Mapping project, a single-page website that poses questions about whether spatial geometry might play an active role in quantum interactions. That chapter takes the project on its own stated terms: as a playful, open proposal that asks for critique.

A word on tone before we begin. Quantum physics attracts hype the way a lamp attracts moths. Words like teleportation invite pictures from science fiction, and words like entanglement invite talk of mystical connection. The real physics is more interesting than either, partly because it is so precise. Where this book says something is known, it is known in the ordinary scientific sense: predicted by a well-

tested theory and confirmed by repeated experiment. Where it says something is speculative, it means exactly that.

Chapter 2. The Qubit

Start with something familiar: a light switch. It is either on or off. Computers are built from vast numbers of tiny switches, each holding a bit, a 0 or a 1. All the information on your phone, every photograph and message, is stored as long strings of these bits. A bit is wonderfully simple. You can look at it without disturbing it, you can copy it as often as you like, and it always has a definite value whether or not anyone is looking.

A qubit, short for quantum bit, is the quantum version of a bit, and it breaks all three of those comfortable rules. A qubit can be made from many physical systems: the polarisation of a single photon, the spin of an electron, two energy levels of a trapped ion, or the direction of a tiny current in a superconducting circuit. What they have in common is that when you measure them in a chosen way, you always get one of two answers, which we label 0 and 1. The surprise is what happens before you measure.

Before measurement, a qubit can be in what physicists call a superposition of 0 and 1. It is tempting to say that this means the qubit is “both 0 and 1 at once”, and you will often read that phrase, but it is misleading. A better way to put it is that the qubit is in a definite state, but that state is not one of the two labels we read out. It is a particular combination of them, with two numbers attached that describe how much of each is present and how they are related to each other. Physicists write such a state as $\alpha|0\rangle + \beta|1\rangle$, where the odd brackets are simply a notation for quantum states, and α and β are the two numbers, called amplitudes.

The amplitudes are not probabilities, but they determine probabilities. If you measure the qubit, the chance of getting 0 is the squared size of α , and the chance of getting 1 is the squared size of β . These two chances add up to one, as chances must. If α and β have equal size, you get 0 half the time and 1 half the time. But the amplitudes carry more than just these chances. They are complex numbers, which means each has a size and also a phase, a kind of angle. The relative phase between α and β does not affect the chances of 0 or 1 in a single measurement of

this kind, but it matters enormously when you measure in a different way, or when the qubit interacts with others.

A helpful picture, used in every textbook, is the Bloch sphere. Imagine a globe. Put 0 at the North Pole and 1 at the South Pole. Every possible state of a single, isolated qubit corresponds to a point on the surface of this globe. The latitude tells you the balance between 0 and 1; the longitude tells you the relative phase. A qubit pointing at the equator, at the longitude of Greenwich, say, is an equal superposition with one phase; a qubit pointing at the equator on the opposite side of the world is an equal superposition with the opposite phase. Measuring “0 or 1” is like asking whether the arrow is closer to the North Pole or the South Pole, and getting an answer at random, weighted by how close it is.

You can also measure along a different axis. Instead of asking “North or South?”, you can ask “Greenwich or the opposite side?” For a qubit made of a photon’s polarisation, this is literally the difference between holding a polarising filter upright or tilting it by forty-five degrees. A qubit that gives a perfectly predictable answer along one axis gives a completely random answer along a perpendicular one. This is the heart of the uncertainty principle as it applies to qubits: there is no state that gives definite answers to every possible question.

Now for the rule that matters most for teleportation. When you measure a qubit, you get a single answer, 0 or 1, and afterwards the qubit is left in the state corresponding to that answer. The original amplitudes are gone. One measurement gives you one bit of information, at most, about a state that took two continuous numbers to describe. If you had a single qubit in an unknown state and wanted to find out exactly where on the Bloch sphere it pointed, you could not do it. You could measure it once and learn a little. To learn more you would need many identical copies, measuring each in different ways and building up statistics, which is what experimental physicists call state tomography. With only one copy, the exact state is fundamentally hidden from you.

This is not a limitation of our instruments. It follows from the structure of quantum theory itself, and it has been tested to extraordinary precision. It is also the reason quantum information is so different from classical information. A classical bit can be read without being disturbed; a qubit generally cannot. A classical bit has a definite value; a qubit in superposition has a definite state, but not a definite value for every question you might ask it.

It is worth pausing on what the amplitudes are not. They are not hidden coin tosses that have already been decided and are simply waiting to be revealed. That interpretation, that the qubit secretly “really is” 0 or 1 and we just do not know

which, can be ruled out by experiments in which the phases of different paths interfere with each other, adding up in some places and cancelling in others, just as waves on a pond do. Coins do not cancel. Amplitudes do. That interference is the fingerprint of genuine superposition, and it appears in everything from the double-slit experiment with single electrons to the operation of quantum computers.

Finally, a word about the word “information”. In quantum physics, information is always carried by something physical: a photon, an atom, an electron, a circuit. When we say that teleportation transfers quantum information, we mean that the state, the particular combination of amplitudes, ends up in a different physical carrier. The carrier at the far end was already there. What changes is the state it is in. Keep that distinction in mind, between the carrier and the state it carries, because it is the key to understanding everything that follows.

Chapter 3. Entanglement

One qubit is interesting. Two qubits can be astonishing. When two qubits interact, they can end up in a joint state that cannot be described by saying what each one is doing separately. Such a state is called entangled, and it is the central resource of quantum teleportation.

The simplest example is a state in which two qubits are either both 0 or both 1, in equal superposition. Physicists write it as $(|00\rangle + |11\rangle)/\sqrt{2}$, and it is one of four special states called Bell states, after the physicist John Bell. If you measure the first qubit and get 0, the second will certainly be 0 too; if you get 1, the second will be 1. Each result on its own is completely random, a perfect coin toss, but the two results always agree.

So far this sounds like a pair of gloves. If I put one glove in each of two boxes and send one box to Edinburgh, then whoever opens the box in Halifax and finds a left glove knows instantly that the Edinburgh box holds a right glove. Nothing mysterious happens; the gloves were left and right all along. The deep question, raised by Albert Einstein, Boris Podolsky and Nathan Rosen in a famous paper of 1935, is whether entanglement is just like the gloves, with the answers fixed in advance and merely unknown, or whether something stranger is going on.

The difference shows up when you measure along different axes. Remember that a qubit can be asked different questions, like a polarising filter held at different angles. For the entangled pair, if both sides measure along the same axis, whatever that axis is, they always agree. That is already hard for the glove picture: the gloves would need to carry pre-arranged answers for every possible angle. In 1964, John Bell showed that this is not merely awkward but impossible to square with the statistics. If each particle carried its own set of instructions, fixed at the source and unaffected by what happens far away, then the correlations between measurements at different angles must obey a certain limit, now called a Bell inequality. Quantum theory predicts that entangled pairs violate that limit.

The first experiments that seriously tested this were done in the 1970s, and a

landmark series was carried out by Alain Aspect and colleagues in Orsay, near Paris, in the early 1980s. In one of these, published in 1982 with Jean Dalibard and Gérard Roger, the measurement settings were switched while the photons were in flight, so that a signal travelling at the speed of light could not have told one side what the other was doing. The results agreed with quantum theory and violated Bell's inequality. Later experiments closed the remaining loopholes. In 2015, a team in Delft led by Ronald Hanson performed a Bell test with electron spins in diamond separated by about 1.3 kilometres, closing the main loopholes at once; other groups in Vienna and Boulder reported loophole-free tests with photons the same year. In 2022, Alain Aspect, John Clauser and Anton Zeilinger shared the Nobel Prize in Physics for experiments with entangled photons that established the violation of Bell inequalities and pioneered quantum information science.

What do these experiments tell us? They do not tell us that one particle sends a signal to the other. They tell us that the correlations between entangled particles cannot be explained by any theory in which each particle carries its own pre-set answers and is influenced only by its immediate surroundings. Physicists describe this by saying that quantum mechanics is non-local, though the word needs careful handling. What it does not mean is that you can use entanglement to send a message.

This point is so important that it is worth spelling out. Suppose Alice in Halifax and Bob in Edinburgh share many entangled pairs. Alice measures her qubits. Each result is random. Bob measures his. Each of his results is also random. Only when they compare notes, by telephone or email or any ordinary method, do they discover that their results are correlated. Whatever Alice chooses to do with her qubits, whichever angle she measures at, or whether she measures at all, the statistics of Bob's results on his own are exactly the same: a fair coin toss. Bob cannot tell, from his side alone, what Alice did. This is called the no-signalling property, and it follows directly from the mathematics of quantum theory. Entanglement produces correlation, not communication.

It helps to think of entanglement as a relationship rather than a property. In the Bell state above, neither qubit has a definite value of its own. What is definite is the relation between them: "same". If you looked at only one of the qubits, ignoring the other, you would find it in what physicists call a maximally mixed state, completely random in every direction, with no information at all. All the information is in the relation. This is one of the most striking features of quantum theory: you can know everything about a whole while knowing nothing about its parts.

That relational character matters later in this book, because a map, too, is a record

of relations. But it is also important to be clear about what entanglement does not do. It does not record a position. The two qubits of a Bell pair do not “know” where each other are. Their correlation is the same whether they are a millimetre apart or a thousand kilometres apart, and it is the same in any direction. Entanglement on its own is not a ruler or a compass. It is a correlation between two records that becomes visible only when those records are brought together and compared.

Entanglement is also fragile. When an entangled qubit interacts with its surroundings, with a stray photon, a vibrating atom or a fluctuating magnetic field, the entanglement leaks into the environment. The two original qubits become less entangled with each other, because each is now partly entangled with countless other particles. This process, called decoherence, is the chief practical obstacle to everything in this book, and we return to it in Chapter 7.

Finally, entanglement can be used up. When it is spent in a protocol like teleportation, it is gone, and a fresh entangled pair must be prepared for the next run. In quantum information science, entanglement is treated rather like a fuel: a resource that can be created, stored for a while, distributed, consumed and measured. Much of the engineering of quantum networks is about producing and distributing that fuel reliably.

Chapter 4. The Copying Ban

If you want to send someone a photograph, the easiest thing is to copy it. You keep the original and send a copy. Classical information is endlessly copyable, and the whole digital world depends on that. It might seem that the simplest way to send a quantum state from Halifax to Edinburgh would be the same: make a copy of the qubit and post it. Quantum theory forbids this.

The no-cloning theorem, published in 1982 by William Wootters and Wojciech Zurek, and independently by Dennis Dieks, says that there is no physical process that can take an arbitrary unknown quantum state and produce two perfect copies of it. You can copy a qubit if you already know it is either 0 or 1, because then you are effectively copying a classical bit. You can prepare as many qubits as you like in a state you know. What you cannot do is build a machine that will faithfully duplicate whatever unknown state is fed into it.

The reason is surprisingly simple, and it comes from a basic feature of quantum theory called linearity. Roughly, the rules of quantum mechanics say that if a process does one thing to 0 and another thing to 1, then what it does to a superposition of 0 and 1 is fixed by those two cases. Suppose you had a copier that turned 0 into “0 and 0” and 1 into “1 and 1”. By linearity, a superposition of 0 and 1 would be turned into a superposition of “0 and 0” with “1 and 1”. That is an entangled state, the very Bell state of the last chapter, and not two independent copies of the original superposition. The copier cannot do both jobs at once. Any machine that copies the two labels correctly will fail to copy the superpositions, and vice versa.

No-cloning ties together several things we have already met. If you could copy an unknown qubit, you could make a thousand copies, measure them all in different ways, and find out exactly what the state was. That would contradict the principle that a single unknown quantum state cannot be fully read. And if you could copy an entangled qubit, you could use the copies to tell which axis a distant partner had measured along, which would allow faster-than-light signalling. The impossibility

of cloning, the impossibility of fully reading a single state, and the impossibility of signalling with entanglement are three faces of the same underlying structure.

No-cloning has practical consequences. It is the basis of quantum key distribution, a way of sharing secret keys in which any eavesdropper who tries to copy the transmitted qubits inevitably disturbs them and reveals themselves. The first such scheme, published by Charles Bennett and Gilles Brassard in 1984, relies precisely on this. Quantum key distribution is now a commercial technology, though it has its own engineering limitations.

For our purposes, the key consequence is this. If you want a qubit in Edinburgh to end up in the same unknown state as a qubit in Halifax, you cannot measure the Halifax qubit and send the results, because measurement does not reveal the state. And you cannot copy the Halifax qubit and send the copy, because copying is forbidden. You could, of course, put the Halifax qubit itself on a plane, or send the photon down an optical fibre, but that is just transporting the physical carrier, and over long distances the qubit is likely to be disturbed on the way.

Teleportation is the clever way around this. It never copies the state, and it never reads the state. Instead, it uses entanglement to move the state from one carrier to another, destroying it at the source as it appears at the destination. At no moment do two copies exist. No-cloning is respected, not evaded. In fact, you can read the teleportation protocol as a demonstration of exactly how much you have to give up, entanglement consumed and two classical bits sent, to achieve what copying would have done for free in a classical world.

Chapter 5. How Teleportation Actually Works

The quantum teleportation protocol was published in 1993 by Charles Bennett, Gilles Brassard, Claude Crépeau, Richard Jozsa, Asher Peres and William Wootters, in a paper with the descriptive title “Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels”. That title is a good summary of the whole thing. There are two channels: a classical one, for ordinary messages, and an “Einstein-Podolsky-Rosen” one, meaning a shared entangled pair. Together they allow an unknown quantum state to be transferred. Neither channel can do it alone.

Let us go through it slowly, using the traditional cast. Alice is in Halifax; Bob is in Edinburgh.

What they start with. Alice has a qubit, which we shall call the payload, in some state $\alpha|0\rangle + \beta|1\rangle$. Crucially, Alice does not need to know what α and β are. The state might have come from someone else, or from a quantum computation; the protocol works for any state. In addition, Alice and Bob share an entangled pair, prepared earlier and distributed to them by some means, perhaps by sending photons through fibre or from a satellite. Alice holds one half of the pair; Bob holds the other. The pair is in the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$.

So there are three qubits in total: the payload and Alice’s half of the pair, both in Halifax, and Bob’s half, in Edinburgh. Bob’s qubit, on its own, is completely random; it carries no information about the payload. It has never interacted with the payload, and it is more than four thousand kilometres away.

Step one: the Bell measurement. Alice takes her two qubits, the payload and her half of the entangled pair, and performs a joint measurement on them. This is not a measurement of each qubit separately. It is a measurement that asks which of the four Bell states the two qubits are in together. Those four states are the one

we have already met, $(|00\rangle + |11\rangle)/\sqrt{2}$, its partner with a minus sign, $(|00\rangle - |11\rangle)/\sqrt{2}$, and two more in which the qubits are opposite rather than the same, $(|01\rangle + |10\rangle)/\sqrt{2}$ and $(|01\rangle - |10\rangle)/\sqrt{2}$. Together they form a complete set of questions you can ask about two qubits, a different set from “is each one 0 or 1?”, and this is called measuring in the Bell basis.

Here is where the mathematics does something beautiful. If you take the three-qubit state and rewrite it so that Alice’s two qubits are described in terms of the Bell states, you find that it splits neatly into four pieces, each with equal weight. In each piece, Alice’s two qubits are in one of the four Bell states, and Bob’s qubit is in a state closely related to the original payload. Specifically, if Alice’s qubits are found in the first Bell state, Bob’s qubit is already in exactly the state $\alpha|0\rangle + \beta|1\rangle$. If they are found in the second, Bob’s qubit is in $\alpha|0\rangle - \beta|1\rangle$, the original with a sign flipped. In the third case it is $\alpha|1\rangle + \beta|0\rangle$, with 0 and 1 swapped, and in the fourth, $\alpha|1\rangle - \beta|0\rangle$, swapped and sign-flipped.

Written out in full, the identity is

$$|\psi\rangle_A |\Phi^+\rangle_{A'B} = \frac{1}{2} \left[|\Phi^+\rangle (\alpha|0\rangle + \beta|1\rangle) + |\Phi^-\rangle (\alpha|0\rangle - \beta|1\rangle) \right. \\ \left. + |\Psi^+\rangle (\alpha|1\rangle + \beta|0\rangle) + |\Psi^-\rangle (\alpha|1\rangle - \beta|0\rangle) \right],$$

where the symbols $\Phi^\pm$ and $\Psi^\pm$ are the conventional names for the four Bell states, the labels A and A' refer to Alice’s payload and her half of the pair, and B is Bob’s qubit. The factor of one half means each of the four outcomes is equally likely. Nothing here is an approximation. It is simply the same state, written in a different way, a bit like rewriting a sum of money in a different currency.

When Alice performs the Bell measurement, she gets one of the four outcomes, entirely at random, each with probability one quarter. Her result tells her nothing whatever about α and β ; it is pure chance. The payload’s original state is destroyed in her laboratory, because her two qubits are now in a Bell state that bears no trace of the amplitudes. And Bob’s qubit, at that instant, is in one of four states, each of which is a simple, known distortion of the original.

Step two: the classical message. Bob does not yet know which of the four states his qubit is in. From his side, nothing has changed: if he were to examine his qubit now, it would look completely random, exactly as before. That is because, averaged over Alice’s four equally likely outcomes, which he does not know, his qubit is still in the maximally mixed state. This is no-signalling in action. Alice must tell Bob which outcome she got. Four outcomes can be named with two classical bits: 00, 01, 10 or 11. She sends those two bits to Bob by any ordinary means, a phone call,

an email, a radio signal or a pulse of light down a fibre. That message can travel no faster than light.

Step three: the correction. When Bob receives the two bits, he knows which of the four states his qubit is in, and he applies the matching correction. If Alice got the first outcome, he does nothing. If she got the second, he flips the sign, an operation called Z. If she got the third, he swaps 0 and 1, an operation called X. If she got the fourth, he applies both: first X, then Z. These operations are simple, fixed rotations of the qubit, the same every time for a given outcome, and they do not require Bob to know anything about α and β . After the correction, Bob's qubit is in the state $\alpha|0\rangle + \beta|1\rangle$, exactly the state the payload had at the beginning.

That is the whole protocol. Let us take stock of what has happened and what has not.

The state has been transferred. Bob's qubit now carries the exact amplitudes that Alice's payload carried, including all their phase information, even though nobody ever learned what those amplitudes were. In principle, with perfect equipment, the transfer is perfect.

No matter has moved. Bob's qubit was in Edinburgh all along. It is the same photon or atom or electron it was before; only its state has changed. Alice's payload is still in Halifax, now part of a Bell pair with her half of the old entangled pair. If the payload were, say, a photon, it might be absorbed by the detector during the measurement, but it does not travel anywhere.

No copy was made. At the moment Alice measured, the original state vanished from her laboratory. It reappeared, after the correction, in Bob's. At no time were there two copies. The no-cloning theorem is satisfied.

Nothing travelled faster than light. Until Bob receives Alice's two bits, his qubit is useless to him; it looks random. The protocol cannot be completed sooner than a light signal could travel from Halifax to Edinburgh. Entanglement made the transfer possible, but the classical message sets its speed limit.

The entanglement was used up. After the protocol, Alice's half of the original pair is entangled with her payload, not with Bob's qubit. To teleport another qubit, they need a fresh entangled pair.

There is a striking asymmetry in the bookkeeping here. Two classical bits, a tiny amount of information, are enough to transfer a state that took two continuous complex numbers to describe, numbers that could in principle encode endless digits. How can two bits carry so much? The answer is that they do not, on their

own. The two bits carry no information about α and β whatever; they are just random. The amplitudes are carried, in a sense, by the correlation in the entangled pair, which was established before the payload even existed. But that correlation is useless without the two bits. Teleportation shows quantum information and classical information working together, each doing a job the other cannot.

It is worth saying a word about the name. Bennett and his co-authors chose “teleportation” with an eye to science fiction, and the word has done a great deal to make the protocol famous. It has also caused a great deal of confusion. A better, if duller, description would be “state transfer using shared entanglement and classical communication”. Physicists often use the term in exactly this narrow sense, and a related protocol, entanglement swapping, uses the same trick to entangle two particles that have never met, by teleporting the entanglement of one pair onto another. Teleportation is also a building block in quantum computing, where qubits are teleported between parts of a processor, and where a variant called gate teleportation is used to perform operations.

Chapter 6. Teleportation in the Laboratory

Theory is one thing; doing it is another. The 1993 protocol is elegant on paper, but each step poses engineering challenges. You need a source of entangled pairs, a way to get them to two places without spoiling the entanglement, a device that can perform a Bell measurement, a fast classical link, and a way for Bob to apply corrections quickly enough to be useful.

The first experimental demonstration was published in 1997 by Dik Bouwmeester, Jian-Wei Pan, Klaus Mattle, Manfred Eibl, Harald Weinfurter and Anton Zeilinger, working in Innsbruck. They used photons, with the qubit encoded in polarisation. Entangled photon pairs were produced by shining an ultraviolet laser into a special crystal, a process called parametric down-conversion, in which one high-energy photon occasionally splits into two lower-energy photons whose polarisations are entangled. The Bell measurement was done by bringing the payload photon and Alice's entangled photon together on a beam-splitter, a partially silvered mirror, and watching for both detectors on the far side to click at the same time.

That Bell measurement was only partial. With simple linear optical components like beam-splitters, you cannot reliably distinguish all four Bell states. The Innsbruck experiment could identify only one of them, the one with opposite polarisations and a minus sign, which occurs a quarter of the time. In that case, Bob's photon is in a state that requires a known fixed correction, and the experiment checked that it matched the payload. The other three quarters of the time, the attempt was simply not counted. This was a genuine demonstration of the teleportation principle, but not a complete, deterministic implementation. Around the same time, a group in Rome led by Francesco De Martini published a related demonstration with a different scheme, and in 1998 a team at Caltech led by Akira Furusawa and Jeff Kimble demonstrated teleportation of a different kind of quantum state, a so-called continuous-variable state of light, in which the

information is carried by the amplitude and phase of a light beam rather than by single photons.

Over the following decades, teleportation was demonstrated in many physical systems: between trapped ions, between atoms in separate laboratories, in superconducting circuits, and between photons and matter. Experiments with trapped ions, which can be held still and measured with high reliability, achieved a fully deterministic Bell measurement, distinguishing all four outcomes and completing the protocol every time rather than a quarter of the time. Distances grew, from a laboratory bench to across cities through optical fibre, and across open air between islands.

The most dramatic distance record came from the Chinese satellite Micius, launched in 2016 for quantum experiments. In 2017, a team led by Jian-Wei Pan, who had been a co-author on the original Innsbruck paper, reported teleporting the states of single photons from a ground station in Ngari, Tibet, to the satellite in orbit, over distances of up to about 1,400 kilometres. The satellite was useful because photons travelling through the vacuum of space, and through only a thin layer of atmosphere when the station is at high altitude, are lost far less than photons travelling the same distance through optical fibre. The same satellite was also used to distribute entangled photon pairs to ground stations more than a thousand kilometres apart.

These experiments are impressive, but it helps to be clear about what the numbers mean. The teleportation in these experiments was not perfect. Physicists measure success with a number called fidelity, which describes how close the state Bob ends up with is to the state Alice started with. A fidelity of 1 means a perfect copy; a fidelity of one half means you might as well have guessed. There is an important benchmark in between: without any entanglement, the best you can do by having Alice measure the payload and send Bob instructions for preparing a new qubit gives an average fidelity of two thirds for a single qubit. To claim genuine quantum teleportation, an experiment must beat that two-thirds threshold convincingly. The satellite experiment reported an average fidelity of around 0.8, which is well above the classical limit but well below perfect.

The experiments also illustrate the role of the classical channel in a very concrete way. In each case, information about the Bell measurement outcome had to be sent by ordinary means to the receiving end, and in many photonic experiments the correction is applied afterwards, or the data are sorted afterwards according to the outcome, rather than in real time. Doing the correction in real time, quickly enough for the teleported qubit to be used immediately, is harder, and has been a

focus of more recent work aimed at building quantum networks.

It is also worth noting the scale of what was teleported. In every experiment so far, the state transferred has been the state of a single qubit, or a small number of qubits, or a mode of a light beam. Some experiments have teleported more complex states, such as several properties of a single photon at once, or states of small collections of atoms. But the gap between teleporting the polarisation of a photon and teleporting anything like a macroscopic object is not a matter of better engineering. It is a gulf of many orders of magnitude, for reasons we turn to now.

Chapter 7. Limits: Why This Is Not a Transporter

Every few years a headline announces that scientists have “teleported” something, and somewhere in the article a sentence wonders whether people might one day travel this way. It is worth taking that question seriously enough to answer it properly, because the answer reveals the real limits of the technology, and they are interesting in their own right.

The classical channel cannot be skipped. As we saw in Chapter 5, teleportation needs two classical bits for every qubit transferred, and those bits must travel by ordinary means, no faster than light. Without them, Bob’s half of the entangled pair is useless; it looks exactly like random noise. This is not a technical inconvenience that a cleverer scheme might remove. It is guaranteed by the no-signalling property of quantum theory, which has been tested in every Bell experiment ever performed. Teleportation is, at best, exactly as fast as light. In practice, with the time taken for measurement, transmission and correction, it is slower.

Entanglement must be distributed in advance, and it degrades. Before Alice can teleport anything, she and Bob must already share entangled pairs, and those pairs must have been delivered by physically sending particles, usually photons, between them or from a common source. That delivery is itself limited by the speed of light and, more severely, by losses. Photons sent through optical fibre are absorbed or scattered at a steady rate, so the fraction surviving falls off exponentially with distance. Over a few hundred kilometres of fibre, only a tiny fraction get through. Amplifiers, which boost classical signals in ordinary telecommunications, cannot be used, because amplifying an unknown quantum state would be a form of copying, forbidden by no-cloning. Physicists have proposed quantum repeaters, which would use entanglement swapping and the storage of qubits in quantum memories to extend entanglement over long

distances in stages, but building practical repeaters remains a major research challenge.

Decoherence. Quantum states are extraordinarily delicate. A qubit in superposition, or a pair in an entangled state, interacts constantly with its surroundings: stray light, thermal vibrations, electromagnetic noise, the gas molecules that remain in even the best vacuum. Each interaction carries away a little information about the qubit into the environment, and as that happens, the superposition or entanglement fades into an ordinary statistical mixture. This process, decoherence, is a central topic of modern quantum physics, studied in detail by Wojciech Zurek and others. It explains why we do not see everyday objects in superpositions: a cat, a chair or a person is constantly interacting with trillions of air molecules and photons, and any superposition of distinct positions would be destroyed in a vanishingly small fraction of a second. For qubits in a laboratory, decoherence times range from microseconds to, in the best cases, seconds or longer, depending on the system and how carefully it is isolated. Every quantum technology is a race against decoherence.

Fidelity is never perfect. Real Bell measurements are imperfect; real entangled pairs are not perfectly entangled; real corrections are slightly off. The result is that teleported states are close to, but not exactly the same as, the originals. For a single qubit, fidelities in the best laboratory experiments are high, but every imperfection accumulates as systems grow more complex. Errors can in principle be corrected using quantum error correction, which spreads one logical qubit across many physical qubits, but this requires substantial overheads and is one of the main engineering frontiers of quantum computing.

The numbers for anything large are absurd. Suppose, for the sake of argument, you wanted to teleport the full quantum state of a small object, say a grain of sand. You would need to perform a joint Bell-type measurement on every one of its roughly hundred billion billion atoms together with the matching halves of an equal number of entangled systems, prepared in advance at the destination. You would need to send at least two classical bits for every quantum degree of freedom. You would need an equal amount of suitable raw material at the far end, already entangled with the source, and every bit of it would have to be protected from decoherence for the duration, which for an object at room temperature is impossible by an almost unimaginable margin. And you would destroy the original state in the process, as no-cloning requires. There is no known physics that forbids this in principle, but there is no plausible route to it either. Teleportation of people, with all the philosophical puzzles that would raise about identity, is not on any realistic horizon.

What teleportation is genuinely good for. None of this makes teleportation unimportant. Its value lies in quantum information processing, not transport. In a future quantum network, teleportation is the natural way to move qubits between distant quantum processors or memories without sending fragile photons directly. It allows a quantum state to be moved from a system good at travelling, such as a photon, to one good at storing, such as an atom or ion. Within a quantum computer, it allows qubits in different parts of the machine to interact. Entanglement swapping, its close cousin, is the core of proposed quantum repeaters. And teleportation has been a superb tool for testing the foundations of quantum theory, because it demands that every subtle feature, entanglement, no-cloning, no-signalling, behave exactly as predicted.

The honest summary is that quantum teleportation is a protocol for relocating quantum states between pre-existing carriers, using up entanglement and requiring ordinary communication. It is fundamental to quantum technology, it has been demonstrated many times over, and it has nothing whatever to do with matter transmission.

Chapter 8. Mapping Space With Quantum Information

We now turn from moving states to measuring space. Here the established physics is rich, but it is spread across several different fields, and it is worth sorting it out carefully before reaching any speculation.

Quantum systems as measuring instruments. The first and most concrete sense in which quantum information can map space is through quantum sensing, also called quantum metrology. Quantum systems are exquisitely sensitive to their surroundings, which is a curse for quantum computing, because it causes decoherence, but a blessing for measurement. An atom's energy levels shift in a magnetic field; the phase of a matter wave shifts in a gravitational field; the frequency of an atomic transition depends on the local gravitational potential, as Einstein's theory of relativity predicts. By measuring these effects carefully, physicists can map fields across space with remarkable precision.

A classic example is neutron interferometry. In 1975, Roberto Colella, Albert Overhauser and Samuel Werner sent neutrons along two paths at different heights in a crystal interferometer and showed that the Earth's gravity shifted the relative phase of the neutron's quantum wave in exactly the way quantum mechanics predicted. That was an early demonstration that the geometry of a gravitational field, a property of space and time around the Earth, leaves a measurable imprint on quantum phases. Modern atom interferometers, in which clouds of ultracold atoms are split and recombined using laser pulses, work on the same principle and serve as sensitive gravimeters and gradiometers, capable of detecting tiny variations in local gravity caused by density differences underground. Optical atomic clocks have become so precise that they can detect the difference in the rate of time between two heights separated by a few centimetres, a direct consequence of general relativity. In a quite literal sense, such clocks can map the shape of the Earth's gravitational field.

Entanglement can improve these measurements. When many particles are measured independently, the precision improves with the square root of their number, a limit often called the standard quantum limit. Using carefully entangled states, it is possible in principle to do better, approaching a limit in which precision improves in proportion to the number of particles, called the Heisenberg limit. Reviews by Vittorio Giovannetti, Seth Lloyd and Lorenzo Maccone set out these ideas, and squeezed light, a related quantum technique, is now used in the gravitational-wave detectors that measure ripples in the geometry of space-time itself. In practice, decoherence erodes much of the advantage of entanglement, and how much gain is achievable depends heavily on the details.

Reference frames. A second, subtler sense concerns how two parties agree about space at all. Every measurement of a qubit is made relative to something. When we say a photon is horizontally polarised, we mean horizontal relative to the laboratory bench. When Bob applies a correction in the teleportation protocol, he rotates his qubit in a way that only makes sense if he and Alice agree what “0” and “1” mean, which in many physical systems amounts to agreeing on a direction in space, or on the setting of a clock.

This is easy to overlook, because in the textbook protocol Alice and Bob are simply assumed to share a common reference frame. But suppose they do not. Suppose Bob’s laboratory is rotated relative to Alice’s by an unknown angle. Then the state he reconstructs will be rotated by that angle too, and from his point of view the teleportation will appear to have failed. Sharing a reference frame is itself a resource, and like entanglement it must be established by physically sending something, a gyroscope, a light signal, a set of qubits pointing in known directions. A substantial body of research, reviewed by Stephen Bartlett, Terry Rudolph and Robert Spekkens in 2007, studies exactly this: how reference frames can be communicated with quantum systems, how much information that takes, and how quantum information can be encoded in ways that do not depend on a shared frame at all.

This is where teleportation and mapping first genuinely meet. Teleportation, as a protocol, is indifferent to distance and direction: the entanglement does not care how far apart Alice and Bob are or in what direction. But the use of teleportation depends on Alice and Bob sharing a frame, which is a spatial fact about their relationship. You can think of the shared frame as a kind of minimal map: an agreement about orientation that has to exist before quantum states can be meaningfully compared between places.

More recent work has gone further, asking what happens when the reference

frames themselves are treated as quantum systems, which can be in superpositions. A 2019 paper by Flaminia Giacomini, Esteban Castro-Ruiz and Časlav Brukner developed a framework for such quantum reference frames, in which whether a system is in a superposition of locations can depend on which reference frame you describe it from. This is active theoretical research, consistent with standard quantum mechanics, and it suggests that “where” something is may be a more relational notion in quantum theory than in everyday life.

What can and cannot be mapped. It is important to be clear about the limits. Entanglement alone cannot measure distance or direction. As we saw, the correlations of a Bell pair are the same whatever the separation, and they cannot be used to send a signal, so they cannot be used to tell you, from one end, anything about where the other end is. To learn spatial facts, you need physical interactions that depend on spatial facts: fields that vary with position, signals that take time to travel, clocks that run at different rates at different heights. Quantum systems can make such measurements more sensitive, and entanglement can help to correlate measurements made in different places, as in proposals for networks of entangled clocks or telescopes linked by shared entanglement. But the spatial information always enters through physics that depends on location, never through entanglement alone.

There is also a limit to how much a single quantum system can tell you. The Holevo bound, proved by Alexander Holevo in 1973, says that a single qubit can carry at most one bit of retrievable classical information, however cleverly you encode it. So a quantum map, in any sense that a person could read, is still made of a finite number of classical readings. The quantum part lies in how those readings are obtained and correlated.

Speculative: space from entanglement. The third sense of mapping is the most ambitious, and it belongs firmly in the category of active theoretical research rather than established fact. Since the late 1990s, string theorists have studied a mathematical correspondence, often called holography or AdS/CFT, between certain theories of gravity in a curved space and certain quantum theories without gravity that live on its boundary. In 2006, Shinsei Ryu and Tadashi Takayanagi proposed a formula relating the amount of entanglement between regions of the boundary theory to the area of surfaces in the curved interior. In 2010, Mark Van Raamsdonk argued that, in this framework, reducing the entanglement between parts of the boundary theory seems to pull the corresponding regions of space apart, and suggested that the connectedness of space might in some sense be built from entanglement. Juan Maldacena and Leonard Susskind’s 2013 proposal, nicknamed ER = EPR, conjectured a deep link between entangled particles and

wormhole-like connections in space-time.

These ideas are taken seriously by many theorists, and they are mathematically precise within the model universes where they are studied. But those model universes differ from our own in important respects, and there is no experimental evidence that space in our universe is built from entanglement. If the reader takes one thing from this paragraph, let it be that “space emerges from entanglement” is a research programme, not a finding.

Chapter 9. Where the Two Ideas Meet

We have now looked at teleportation as a protocol and at mapping as a family of ideas. It is time to bring them together, being careful to mark where established physics ends.

Common ground. The first meeting point is entanglement itself. In teleportation it is a resource that is spent; in quantum-enhanced sensing it is a resource that sharpens measurements; in the speculative holographic picture it is proposed as something like the fabric of geometry. In all three cases, entanglement is a relation between systems, and in none of them does it, on its own, carry a message or a measurement from one place to another.

The second meeting point is the classical channel. Teleportation cannot finish without two classical bits. A distributed sensor network cannot produce a map until readings are brought together and compared, which also requires classical communication. In both cases, quantum correlations sit in the background, waiting to be made useful by ordinary information travelling at or below the speed of light. The universe seems to insist that whatever quantum correlations may exist, the creation of a usable record of them goes through the slow, local, classical world.

The third meeting point is the reference frame. Teleportation presumes that Alice and Bob agree on what their qubit states mean. Mapping is, almost by definition, the business of establishing such agreements: which way is north, how long a metre is, when noon falls. A quantum network that teleports qubits between distant nodes must also maintain shared frames between those nodes, by synchronising clocks, aligning polarisation axes, and compensating for the twisting of photon polarisation as light travels through fibre. In practice, much of the engineering of real teleportation experiments is precisely this kind of map-keeping.

A useful way of putting it. The formula sheet that accompanies the Quantum Spatial Mapping website, which we shall look at more closely in the next chapter, contains a short note that captures this rather well. It describes the correlation in an entangled pair as “not a pointer to a place” but “a lock between two records”, and adds that “space stays the stage those records sit on”. That is a fair plain-language summary of the established position. Entanglement links the outcomes of measurements; it does not encode where the measured particles are. Space, in standard quantum mechanics, remains the background against which those records are made.

The same note goes on to suggest a direction: to “treat the lock as a relation, then ask what geometry or relocated state that relation can stand in for — after a classical channel is added, if you actually want to use it.” The phrase “relocated state” is a fair description of teleportation: a relation (the shared Bell pair) plus a classical message stands in for a state moved from one place to another. The phrase “what geometry ... that relation can stand in for” points at a more open question, and it is here that we cross into interpretation.

Speculative: what a quantum spatial map could mean. What follows is not established physics. It is an attempt to set out, as clearly and honestly as possible, a few things that the phrase “quantum spatial mapping” could reasonably be taken to mean, and how each one stands.

One modest reading is that a quantum spatial map is a map of space made with quantum instruments: an atlas of gravitational or magnetic fields obtained from networks of atomic clocks, atom interferometers or other quantum sensors, perhaps linked by shared entanglement to improve their combined precision. This reading is entirely grounded. Such maps are being made now in a limited way, and proposals for entangled sensor networks are an active area of research. Nothing in it conflicts with known physics.

A second reading is that a quantum spatial map is a description of space in terms of relations between quantum systems rather than coordinates. Here we can draw on relational ideas that already exist within physics. Don Page and William Wootters showed in 1983 how time can be described in quantum mechanics as a correlation between a clock and the rest of a system, rather than as an external parameter. Quantum reference frame research does something similar for position. One could imagine, in the same spirit, describing the spatial arrangement of a set of quantum systems entirely through the correlations among them, with distance and direction emerging from those correlations rather than being assumed in advance. This is a legitimate theoretical direction. Whether it yields anything new, beyond a

reformulation of standard physics, depends on whether it makes predictions that the standard picture does not.

A third, boldest reading is that space itself is a kind of quantum map: that geometry is a summary of the pattern of entanglement among underlying quantum degrees of freedom, as the holographic research programme suggests in its model universes. If that were true of our universe, teleportation would look rather different. Moving a quantum state across space using entanglement would be, in a sense, a process that happens within the very structure that gives space its shape. This is a beautiful idea, and it is pursued seriously, but it is not established, and nobody yet knows how to test it directly.

A fourth reading, closest to the question the Quantum Spatial Mapping website poses, asks the reverse: not whether entanglement makes geometry, but whether geometry shapes quantum behaviour in ways we have not noticed. Here, again, some of the answer is already known. Geometry certainly affects quantum systems. Gravity shifts quantum phases, as the neutron experiments showed. A quantum particle confined to move on a curved surface feels an effective force that depends on the curvature of the surface, a result that has been studied since the early 1980s. Quantum field theory in curved space-time, the framework in which Stephen Hawking predicted that black holes radiate, describes how curved geometry changes what counts as a particle at all. There are also proposals, associated with Lajos Diósi and Roger Penrose, that gravity might play a role in why quantum superpositions of massive objects do not persist, and experiments have been proposed, by Sougato Bose and colleagues and by Chiara Marletto and Vlatko Vedral in 2017, to test whether gravity can entangle two masses. These are real research questions at the boundary of quantum theory and gravity. None has yet produced a result that overturns the standard picture.

So the honest summary of where teleportation and mapping meet is this. They meet in entanglement as a relational resource, in the indispensability of classical communication, and in the dependence of both on shared reference frames. Beyond that, they meet in a set of open questions about whether space is a passive stage, as standard quantum mechanics treats it, or something more active. Those questions are worth asking. They have not been answered.

Chapter 10. The Quantum Spatial Mapping Project

The Quantum Spatial Mapping project lives at a single web page, www.quantumspatial.lol, backed by a public GitHub repository. At the time of writing, the repository contains the web page itself and a two-page PDF titled “Quantum teleportation formulas”, linked from the site. This chapter describes what those materials say, and then responds to them in the spirit they invite. I have not drawn on any other source about the project, and nothing here should be taken as a claim about results, measurements or publications, because the page itself makes no such claims.

What the page says. The site is headed “Quantum Spatial Mapping”, with the subtitle “Exploring a new branch of quantum physics”, and a logo described in its own code as “an atom with a question mark at its center”. Under the heading “Theory”, it describes the project as “a playful, evolving proposal for thinking about the relationship between space and quantum phenomena.” It continues: “Rather than treating space as a passive container for quantum events, this framework asks whether spatial geometry might play an active role in mediating quantum interactions. The idea is intentionally open to scrutiny, questions, and better explanations.”

A section headed “Mathematical Framework” presents three expressions, introduced as “a starting point for discussion, building on familiar quantum-mechanical notation”. It then asks: “What do these terms mean here? Which assumptions are missing? That is precisely where constructive responses are welcome.”

A section headed “We Want Your Thoughts” describes the project as “an evolving theory” and invites readers to “send a critique, an alternative interpretation, a reference, or simply tell us where the joke stops being funny.” It offers a feedback email address, a link to open an issue on GitHub, and the link to the teleportation

formula sheet. A list of “Questions to Explore” asks: “Could spatial geometry influence quantum measurement?”, “What predictions would distinguish this idea from established models?”, “Where do the mathematics need clarification or correction?” and “Is this physics, speculative fiction, or both?” The footer reads: “Quantum Spatial Mapping · serious questions, questionable certainty”.

How to read it. Taken on its own terms, the page is not a laboratory report or a research paper. It reports no experiments, data or predictions, and it does not claim to. It is better understood as a public prompt: a conceptual, partly playful proposal that sets out a question, sketches some notation, and asks readers to help sharpen it, or to say where it fails. The domain name, the question mark at the centre of the atom, the line about “where the joke stops being funny” and the footer’s “questionable certainty” all signal a light touch. At the same time, the questions it lists are real questions, and some of them, as Chapter 9 showed, connect to active research. The fairest response is to take the questions seriously and the framework as what it says it is, a starting point.

The formula sheet. The accompanying PDF is a clear, correct summary of the standard teleportation protocol, consistent with the account in Chapter 5. It begins with Alice’s unknown qubit $\alpha|0\rangle + \beta|1\rangle$ and a shared Bell pair, lists the four Bell states, and notes that the Bell measurement “has four equally likely outcomes”. It says that Alice “sends the two classical bits that name which of the four outcomes she got”, and states plainly: “Entanglement is consumed. The particle is not moved, and the state is not sent as a signal.” It lists Bob’s four possible states before correction and the matching operations, I, Z, X, and ZX, with the note that “ZX means X first, then Z”, and it writes out the joint state expanded in the Bell basis with its factor of one half. I checked each of these against the standard derivation, and they are right. The sheet refers to “the protocol pages”, which are not present in the repository as I found it; presumably they are planned or exist elsewhere.

The sheet ends with the short note on what it calls “the relation lock”, quoted in Chapter 9: that correlation “is not a pointer to a place” but “a lock between two records”, that “space stays the stage those records sit on”, and that the “usable move for the quantum-spatial approximation” is to treat the lock as a relation and ask what geometry or relocated state it can stand in for, “after a classical channel is added, if you actually want to use it.” This is, to my reading, the most careful statement of the project’s idea anywhere in its materials, and it is notably cautious: it keeps space as the stage, and it insists on the classical channel. Nothing in it conflicts with established physics.

The three expressions. The page explicitly asks where its mathematics need

clarification or correction, so it seems fair to offer some, gently and constructively. The three expressions, as printed on the page, are:

$$H = H_0 + \frac{\hbar^2}{2m} \nabla^2 \Psi + V(r, t)$$

$$T_{\text{spatial}} = \frac{\hbar^2}{2m} \sqrt{\det g} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu \Psi \right)$$

$$U(t) = \exp(-i H_{\text{spatial}} t / \hbar)$$

The first resembles the familiar Hamiltonian, the operator that represents a system's total energy, written as a free part, a kinetic part and a potential. Two small points stand out. In standard notation the kinetic energy operator carries a minus sign, $-\frac{\hbar^2}{2m} \nabla^2$. And the expression mixes an operator with a wavefunction: $\nabla^2 \Psi$ is the result of applying an operator to a state, whereas H_0 and V are operators. Writing $H = H_0 - \frac{\hbar^2}{2m} \nabla^2 + V(r, t)$, and then letting it act on Ψ , would make the intended meaning clearer, and would also raise the question of what H_0 contains that the kinetic and potential terms do not.

The second expression is the most interesting, because it is clearly reaching for something real. The combination of a metric $g^{\mu\nu}$, its determinant, and derivatives arranged in this way is the shape of the Laplace–Beltrami operator, the standard generalisation of ∇^2 to curved spaces. In conventional form it reads $\frac{1}{\sqrt{\det g}} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu \Psi \right)$, with the outer factor dividing rather than multiplying. With that factor inverted, and with the minus sign restored, $-\frac{\hbar^2}{2m}$ times this operator is exactly the kinetic energy of a quantum particle moving in a curved space with metric g . That is established physics: it is how one writes the Schrödinger equation on a sphere, on a curved surface, or in curvilinear coordinates. It is a genuinely good starting point for a framework that asks whether geometry plays an active role, because in this operator the geometry, through g , directly shapes how the wavefunction spreads and interferes. One clarifying question the project might address is whether g here is meant to be fixed, as in ordinary quantum mechanics on a curved background, or dynamical, responding to the quantum state; and whether the indices μ and ν run over space only or over space-time, since the expression is written with a Schrödinger-style $\hbar^2/2m$ but with indices conventionally used for space-time.

The third expression is the standard form of the time-evolution operator, which

tells you how a state changes over time under a given Hamiltonian. It is correct as written for a Hamiltonian that does not itself change with time. Since the first expression includes a potential $V(r, t)$ that depends on time, the simple exponential would need to be replaced by a time-ordered version if H_{spatial} inherits that time dependence. The page also does not yet define H_{spatial} ; presumably it is meant to be built from T_{spatial} and a potential, and stating that explicitly would tie the three expressions together.

Answering the page's questions. The page asks four questions, and it seems right to close the chapter by giving honest, brief answers drawn from the rest of this book.

“Could spatial geometry influence quantum measurement?” Geometry certainly influences quantum evolution: curved-space kinetic terms, gravitational phase shifts and the geometric potentials felt by constrained particles are all established. Whether geometry influences measurement itself, the moment at which a definite outcome appears, is open. Proposals such as the Diósi-Penrose idea of gravity-related collapse say yes in a specific way, and experiments are being designed to test them. A version of Quantum Spatial Mapping that made a definite claim here would be connecting with a live question.

“What predictions would distinguish this idea from established models?” This is the decisive question, and the page is right to ask it. As currently written, with the corrections above, the framework reproduces ordinary quantum mechanics on a curved background, which is well understood. To become a distinct proposal it would need an ingredient that changes a prediction: for example, a rule by which the metric responds to the quantum state, or a term that makes measurement outcomes depend on geometry in a way that standard theory does not. Any such rule would also need to respect what we know to be true: the no-signalling property, the success of Bell-test predictions, and the fact that teleportation needs a classical channel.

“Where do the mathematics need clarification or correction?” The notes above suggest a starting list: the sign of the kinetic term, the separation of operators from states, the placement of the determinant factor, the definition of H_{spatial} and the meaning of the indices.

“Is this physics, speculative fiction, or both?” On the evidence of the page, it is currently a speculative, playful prompt built on real physics. The formula sheet is straight physics, and correct. The curved-space operator is real physics with a small slip. The central proposal, that geometry might actively mediate quantum interactions, is speculation, but speculation in a direction that serious researchers

also explore. Whether it becomes physics depends on whether it ever makes a prediction that could be wrong. That is not a criticism; it is simply how ideas become science, and the page itself appears to understand that.

Closing

We began with two friends, one in Halifax and one in Edinburgh, and the difference between sending a thing, sending a message and sending a description. Quantum physics, it turns out, adds a fourth possibility: sharing a correlation in advance, and later using it, together with a short ordinary message, to move a state from one carrier to another. That is teleportation. It is real, it has been done many times, and it neither moves matter nor outruns light.

Mapping is something else: the patient business of recording relations. Quantum systems can help to make maps more precise, and the question of how distant parties agree about directions and clocks turns out to be inseparable from the question of how quantum information is used. Whether space itself is in some deep sense a map of quantum relations is one of the open questions of modern physics.

The Quantum Spatial Mapping project sits at the playful edge of these ideas. It asks whether space is merely the stage or also one of the actors, and it asks readers to help work out what that would mean. Its own formula sheet gives a sensible answer to part of that question: entanglement is a lock between records, not a pointer to a place, and to use it you still need a classical channel. The rest of the question is open. The useful next step, for this project as for any idea in physics, is to find the place where it predicts something the standard picture does not, and then to see whether nature agrees.

Further Reading

Textbooks and general books

Michael A. Nielsen and Isaac L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000; 10th anniversary edition 2010). The standard textbook; chapter 1 includes a clear account of teleportation.

John S. Bell, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge University Press, 1987). Bell's collected papers on the foundations of quantum theory.

Leonard Susskind and Art Friedman, *Quantum Mechanics: The Theoretical Minimum* (Basic Books, 2014). A gentle mathematical introduction for the general reader.

Nicolas Gisin, *Quantum Chance: Nonlocality, Teleportation and Other Quantum Marvels* (Springer, 2014). A non-technical book on entanglement and teleportation by a leading experimentalist.

John Preskill, lecture notes for Physics 219, Quantum Computation (California Institute of Technology), freely available online.

Foundational papers

A. Einstein, B. Podolsky and N. Rosen, "Can quantum-mechanical description of physical reality be considered complete?", *Physical Review* 47, 777 (1935).

J. S. Bell, "On the Einstein Podolsky Rosen paradox", *Physics* 1, 195 (1964).

W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned", *Nature* 299, 802 (1982). See also D. Dieks, *Physics Letters A* 92, 271 (1982).

C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres and W. K. Wootters, "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels", *Physical Review Letters* 70, 1895 (1993).

A. S. Holevo, “Bounds for the quantity of information transmitted by a quantum communication channel”, *Problems of Information Transmission* 9, 177 (1973).

D. N. Page and W. K. Wootters, “Evolution without evolution: dynamics described by stationary observables”, *Physical Review D* 27, 2885 (1983).

Experiments

A. Aspect, J. Dalibard and G. Roger, “Experimental test of Bell’s inequalities using time-varying analyzers”, *Physical Review Letters* 49, 1804 (1982).

B. Hensen et al., “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres”, *Nature* 526, 682 (2015).

D. Bouwmeester, J.-W. Pan, K. Mattle, M. Eibl, H. Weinfurter and A. Zeilinger, “Experimental quantum teleportation”, *Nature* 390, 575 (1997).

A. Furusawa et al., “Unconditional quantum teleportation”, *Science* 282, 706 (1998).

J.-G. Ren et al., “Ground-to-satellite quantum teleportation”, *Nature* 549, 70 (2017).

R. Colella, A. W. Overhauser and S. A. Werner, “Observation of gravitationally induced quantum interference”, *Physical Review Letters* 34, 1472 (1975).

Reviews and research directions

S. Pirandola, J. Eisert, C. Weedbrook, A. Furusawa and S. L. Braunstein, “Advances in quantum teleportation”, *Nature Photonics* 9, 641 (2015).

W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical”, *Reviews of Modern Physics* 75, 715 (2003).

S. D. Bartlett, T. Rudolph and R. W. Spekkens, “Reference frames, superselection rules, and quantum information”, *Reviews of Modern Physics* 79, 555 (2007).

V. Giovannetti, S. Lloyd and L. Maccone, “Advances in quantum metrology”, *Nature Photonics* 5, 222 (2011).

F. Giacomini, E. Castro-Ruiz and Č. Brukner, “Quantum mechanics and the covariance of physical laws in quantum reference frames”, *Nature Communications* 10, 494 (2019).

S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT”, *Physical Review Letters* 96, 181602 (2006).

M. Van Raamsdonk, “Building up spacetime with quantum entanglement”, *General Relativity and Gravitation* 42, 2323 (2010).

J. Maldacena and L. Susskind, “Cool horizons for entangled black holes”, *Fortschritte der Physik* 61, 781 (2013).

The project

Quantum Spatial Mapping, www.quantumspatial.lol, and its accompanying formula sheet “Quantum teleportation formulas” (PDF linked from the site).